CONFIGURE IPv6 ADDRESSING

Diploma in Cyber Security

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Introduction

IPv6 is the newer version of Internet Protocol used to identify devices on a network. In this practical, I will work with static IPv6 addresses, Unique Local Addresses (ULA), temporary IPv6 addresses, a small restaurant search program, and a basic IPv6 subnetting example.

Task 1 – Assign and Verify a Static IPv6 Address

Open Windows or Linux and identify the primary network adapter. A static IPv6 address can be assigned to the adapter using the operating system network settings or command line. After assigning the address, verify it from the command line.

Windows verification command:

ipconfig

Linux verification command:

ip addr

Example documentation address: 2001:db8:1234:1::10/64. The 2001:db8::/32 range is reserved for documentation, so it should be treated only as an example in this assignment.

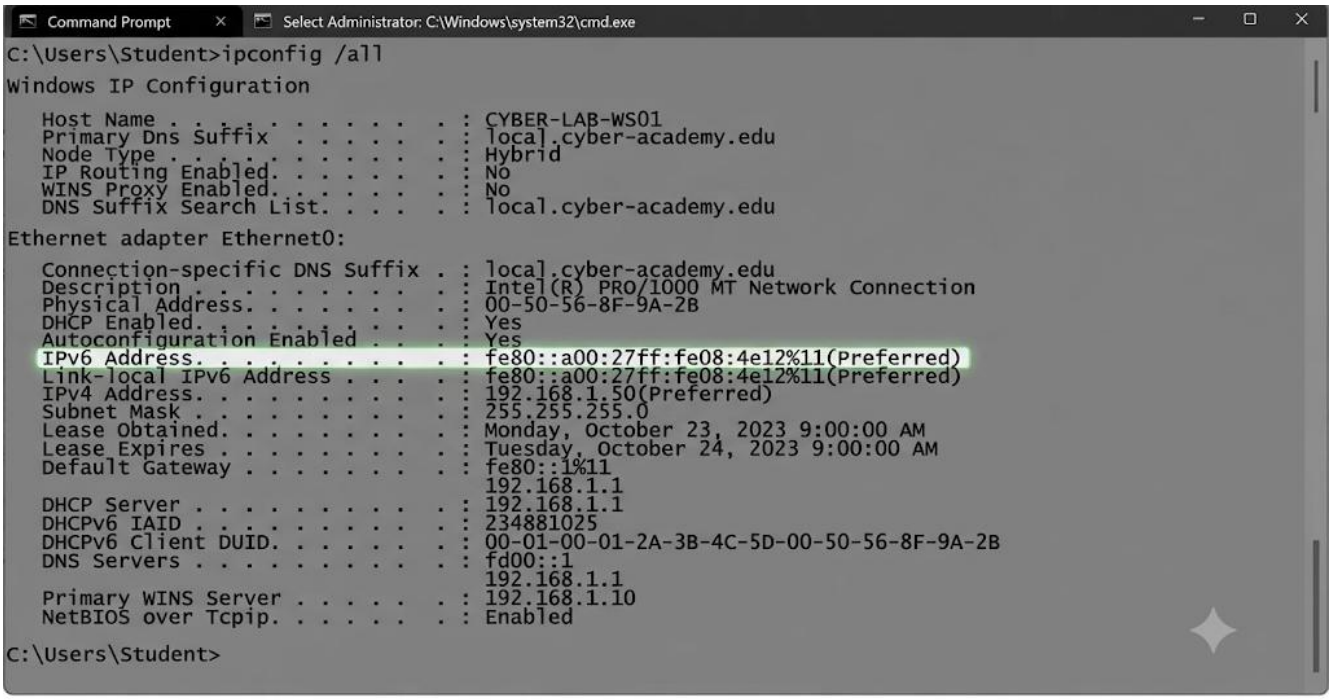


Figure 1: IPv6 Address Verification using ipconfig

FIGURE 1 HERE – IPv6 configuration verification]

Show the command output with the IPv6 address visible.]

Task 2 – Configure a Unique Local IPv6 Address

Unique Local Addresses (ULA) normally start with fd. For the practical, configure a ULA on the system and test communication with another LAN device that also has a ULA.

Example ULA: fd12:3456:789a:1::10/64

Windows IPv6 ping command:

ping -6 fd12:3456:789a:1::20

Linux IPv6 ping command:

ping6 fd12:3456:789a:1::20

If the other device replies, the IPv6 connectivity test is successful. Use the real IPv6 address of the other LAN device during the practical.

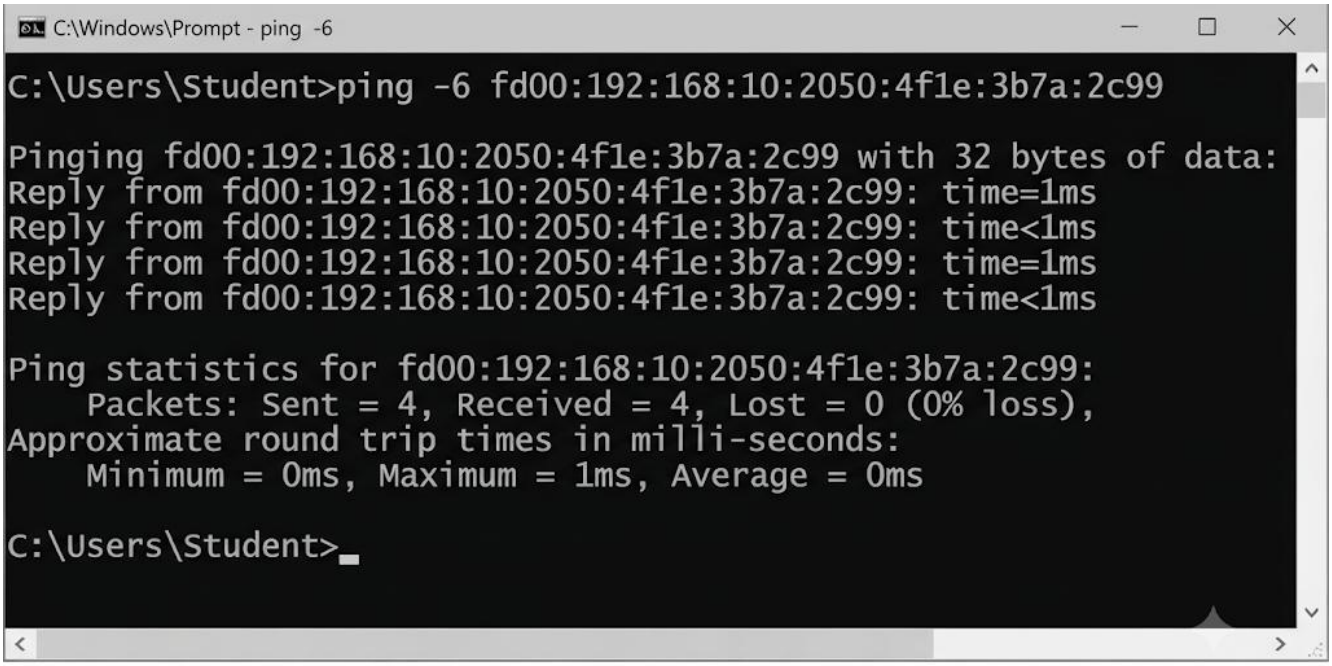


Figure 2: ULA IPv6 Connectivity Test

FIGURE 2 HERE – Successful IPv6 ping test

Show ping -6 or ping6 with successful replies

Task 3 – Temporary IPv6 Address and Reconnection Test

A temporary IPv6 address can be used for privacy on systems that support IPv6 privacy extensions. First record the current IPv6 address, then disconnect/reconnect the network or reboot the system and check the address again.

Suggested Windows command:

ipconfig

Suggested Linux command:

ip addr

Record the actual observed result in a text file.

Stage	IPv6 Address Observed	Observation
Before disconnect/reboot	Enter actual address	Enter actual observation
After reconnect/reboot	Enter actual address	Enter actual observation

Note: The address may or may not change depending on the operating system, network configuration, privacy settings, and how the connection is renewed.



FIGURE 3– IPv6 address before reconnect/reboot

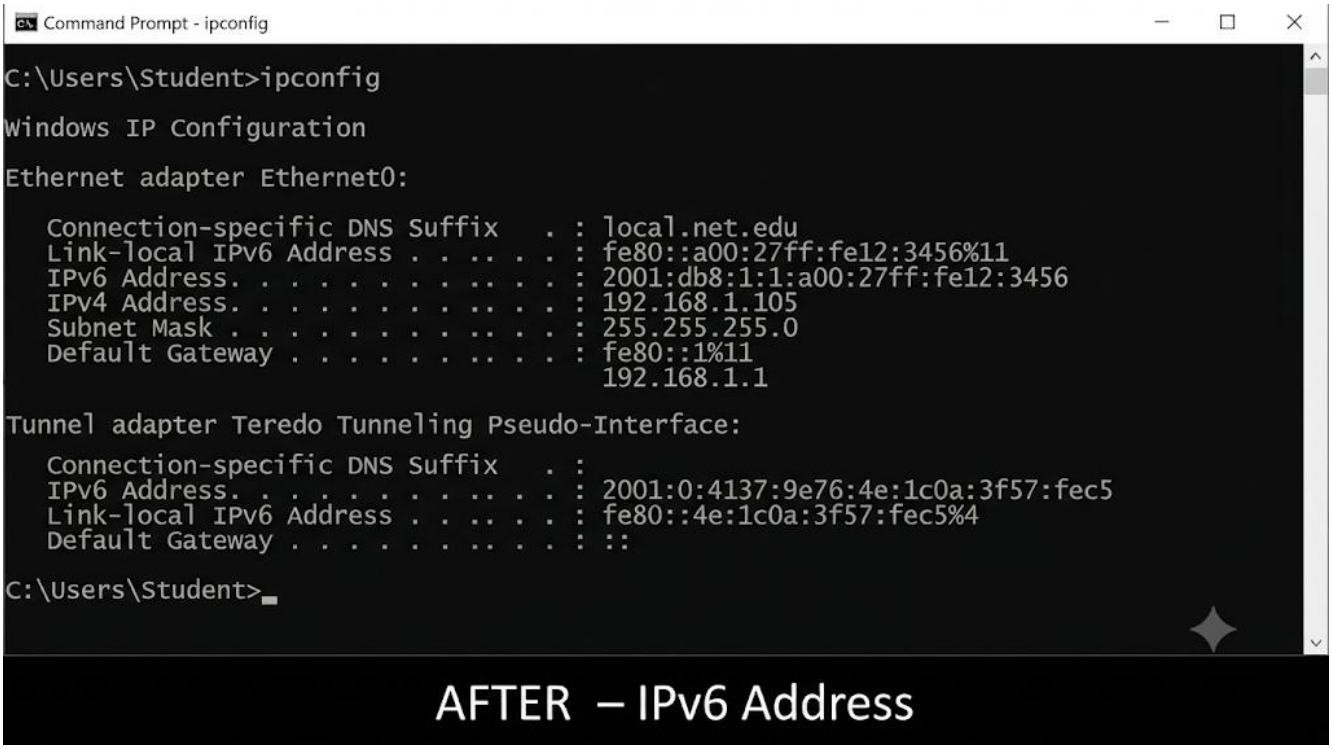


FIGURE 4– IPv6 address after reconnect/reboot

Task 4 – Zomato-Style Restaurant IPv6 Search

For this task, a small Python program stores restaurant names with example IPv6 addresses. The user enters part of a restaurant name and the program displays matching restaurants and their IPv6 addresses.

```
restaurants = {
    "Pizza House": "2001:db8:100::30",
    "Burger Point": "2001:db8:100::31",
    "Spice Kitchen": "2001:db8:100::32",
    "Royal Cafe": "2001:db8:100::33"
}

search = input("Enter restaurant name:
").lower()

for name, ipv6 in restaurants.items():
    if search in name.lower():
        print(name, "->", ipv6)
```

Example output:

> ***Enter restaurant name: pizza***
> ***Pizza House -> 2001:db8:100::30***

The IPv6 addresses in this script are documentation examples, not real restaurant server addresses.

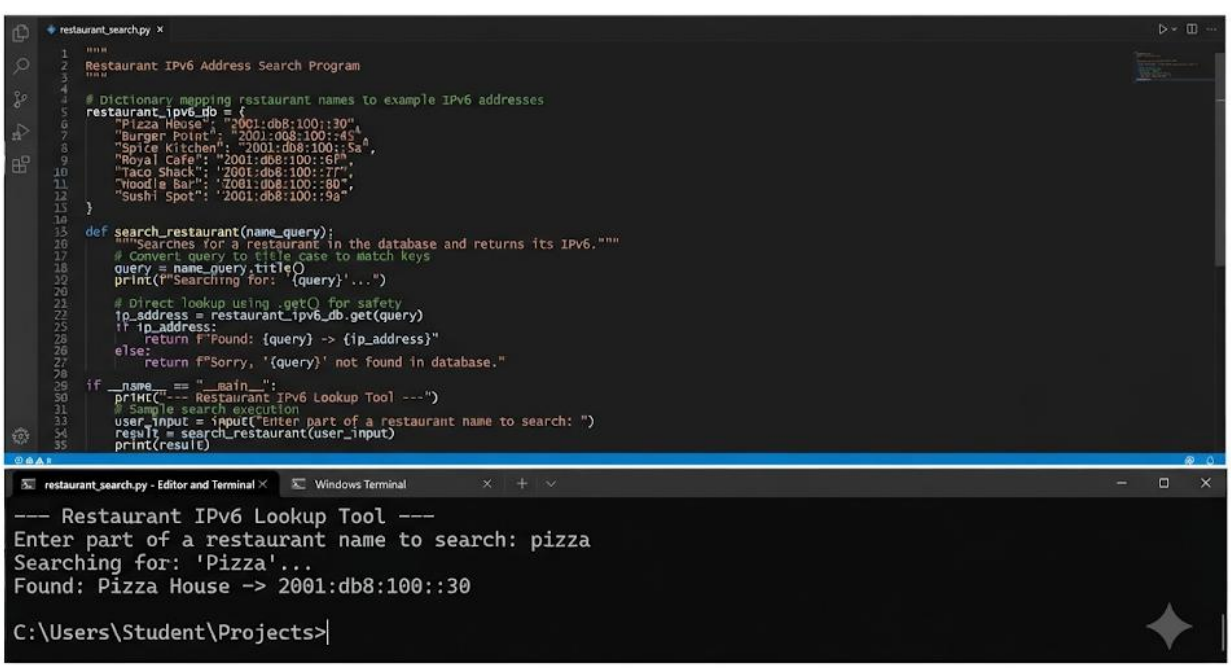


Figure 5: Restaurant IPv6 Search Program

FIGURE 5 – Python restaurant search and output]

Task 5 – IPv6 Subnetting Scenario and Manual Assignment

The following is a sample scenario generated with the help of ChatGPT/Copilot. It uses a documentation IPv6 prefix and creates at least three /64 subnets.

Base prefix: 2001:db8:1234::/48

- Subnet 1: 2001:db8:1234:1::/64
- Subnet 2: 2001:db8:1234:2::/64
- Subnet 3: 2001:db8:1234:3::/64

Example manual address from Subnet 1: 2001:db8:1234:1::10/64

Example configuration steps:

1. Identify the primary network adapter.
2. Choose one valid address from the selected /64 subnet.
3. Assign the address using the operating system's IPv6 network settings or suitable command line tools.
4. Verify the address using ipconfig on Windows or ip addr on Linux.
5. Test IPv6 connectivity if another device is available.

For the actual practical, use an IPv6 prefix and address that are valid for your own lab network. Do not claim the documentation prefix above as a real Internet or lab address unless your instructor specifically uses it.

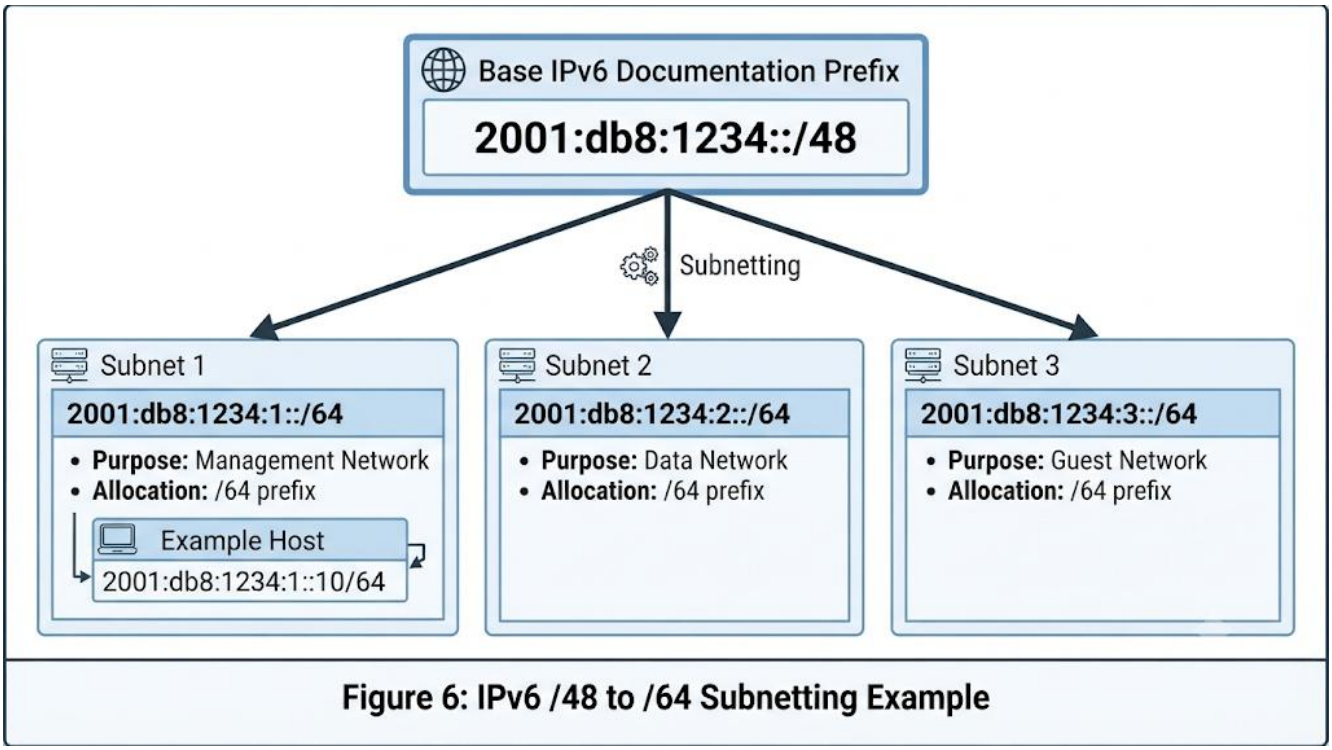


FIGURE 6 – IPv6 subnetting diagram

Conclusion

In this practical, I learned how IPv6 addresses can be assigned and checked, how ULA addresses beginning with fd can be used inside a LAN, how an IPv6 address can behave after reconnecting a network, how to search a small restaurant list using IPv6 values, and how a /48 IPv6 prefix can be divided into /64 subnets.

Quick Revision

- IPv6 uses 128-bit addresses.
- ULA addresses commonly use the fd prefix.
- Windows can verify IPv6 configuration with ipconfig.
- Linux can verify IPv6 configuration with ip addr.
- ping -6 or ping6 can be used for IPv6 connectivity testing.
- A /48 IPv6 prefix can be divided into many /64 subnets.